

# DAY 27 OF 75 DAYS ETHICAL HACKING COURSE



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## DNSSEC

As you would already know, DNS stands for Domain Name System. The DNS protocol is responsible mainly for resolving domain names. Instead of remembering the IP address, you need to focus on the domain name. For instance, at the time of this writing, example.com resolves to 93.184.216.34; it is clear which one is easier for the human mind to remember.

DNS works by sending a DNS query. For instance, when browsing the web, your web browser might send a query for DNS record type A or AAAA, i.e., IPv4 or IPv6 addresses. In the following console output, we can see the host with IP address 192.168.0.102 sending two DNS queries to the DNS server 1.0.0.1 regarding the domain name example.com. We can see the responses for the A and AAAA queries.

This name-to-IP address resolution is very convenient; however, anyone on the network could have responded with a forged response. Furthermore, the host that sent the query would have accepted "any" response. In other words, the host would connect to a rogue server. One way to avoid such a situation would be by using DNSSEC. DNSSEC makes it possible to ensure that the DNS response we receive is from the domain owner. To achieve this, DNSSEC requires two main things:

1. The DNS zone owner should sign all DNS records using their private key.
2. The DNS zone publishes its public key so users can check the validity of the DNS records signatures.

## SSL/TLS Protocol

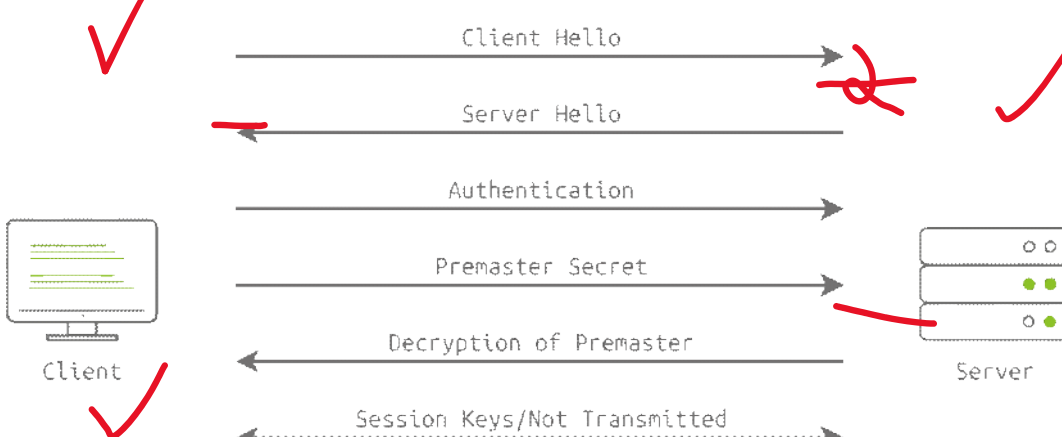
### Technical Overview - SSL/TLS

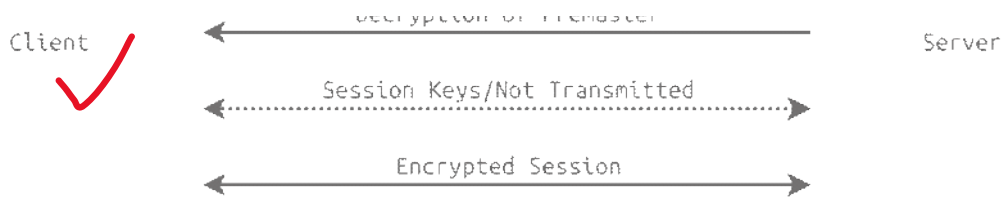
Secure Socket Shell (SSL) and Transport Layer Security (TLS) are protocols used to encrypt data exchanged between a client, such as a web browser, and a server. Consider SSL/TLS as a wrapper that encrypts various communication protocols, such as HTTP and FTP, to create HTTPS and FTPS. SSL is not commonly used nowadays as TLS has been gradually replacing it.

### SSL/TLS Workflow

SSL/TLS handshake is performed to encrypt the communication between client and server through the following steps:

1. Client Hello Message: The client sends a hello message to the server, it includes the client TLS version and the cypher suite that the client supports, in addition to random bytes.
2. Server Hello Message: The server responds with a hello message, highlighting its certificate, chosen cypher suite and random bytes.
3. Authentication: The client authenticates the server's certificate through the certificate authority that issued it. For example, when we visit [Google](#), Google shares its certificate. The received certificate is verified by our browser, which is pre-installed with the certificates of various certificate authorities.
4. Premaster Secret: The client encrypts random bytes with the server's public key. (The client retrieves the public key from the server's certificate.)
5. Decryption of Premaster: The server decrypts the premaster with its private key.
6. Session Keys Generated: The client and the server generate session keys based on client random bytes
7. Ready Messages: The client and server send a "finished" message using the session key to indicate that the session is ready for transmission. The client and server are now ready to exchange messages over SSL/TLS encrypted connection.





TLS is a wrapper that encrypts communication of communication protocols. It has port numbers for various protocols, such as 443 for HTTPS and 990 for FTPS.

1. Server Hello Message
2. Decryption of Premaster
3. Client Hello Message
4. Session Keys Generated:
5. Authentication
6. Ready Messages:
7. Premaster Secret:



## IPsec

IPsec stands for Internet Protocol Security. In this room, we use IPsec to refer to IPsec-v3. IPsec provides security by adding authentication and protecting the integrity and confidentiality of the network traffic. IPsec uses the following protocols:

1. Authentication Header (AH): Provides authentication and integrity.
2. Encapsulating Security Payload (ESP): Provides authentication, integrity, and confidentiality.
3. Security Association (SA): Is responsible for negotiating the encryption keys and algorithms. One example is Internet Key Exchange (IKE). Discussing SA in more detail is outside the scope of this room.

## VPN

When the TCP/IP protocol was designed, security requirements such as confidentiality and integrity were not a design target. In contrast, availability was the priority as one of the purposes of the Internet is to withstand a nuclear attack, as is evident by the routing protocols adapting quickly when a link goes down. But we need to allow a corporation to use the existing Internet infrastructure to connect its offices securely. The answer lies in setting up a VPN.

A Virtual Private Network (VPN) makes it possible to establish a private connection over a public network. In other words, we can establish a secure connection over an insecure infrastructure.

The two most common protocols used to establish VPN connections are:

1. IPsec
2. SSL/TLS

IPsec's ESP is a perfect protocol for setting up secure tunnels between different networks or a computer and a network. ESP can provide security and integrity of all data transmitted between two points; moreover, even the IP address can be hidden in tunnel mode. Note that the system must be behind a VPN concentrator for the IP address to be hidden. Cisco VPN systems offer IPsec.

